

Qualification Pack



Mine Electrician

Electives: Underground Metal/ Opencast/ Underground Coal

Options: Underground Installation

QP Code: MIN/Q3101

Version: 3.0

NSQF Level: 4.0

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MIN/Q3101: Mine Electrician

Brief Job Description

Mine Electrician ensures installation, operation and maintenance (O&M) of the electrical substations, equipment and electrical supply system. The role holder also ensures that all the electrical systems and machinery work is in accordance with relevant specifications for opencast and underground mines (including gassy mines).

Personal Attributes

This job requires quality consciousness, safety orientation, physique to sustain strenuous conditions and dexterity.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N3101: Identify work requirements, related processes and arrange equipment for electrical system](#)
2. [MIN/N3102: Install the electrical systems/sub-stations and equipment](#)
3. [MIN/N3103: Operate and maintain electrical systems/substation and equipment](#)
4. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Electives(*mandatory to select at least one*):

Elective 1: Underground Metal

This unit is about adhering to health, safety and environmental guidelines in Underground Metalliferous Mines (UMM) during execution of various tasks, operations and maintenance.

1. [MIN/N1702: Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines](#)

Elective 2: Opencast

This unit is about adhering to health, safety and environmental guidelines in the Open Cast Mines during execution of various tasks, operations and maintenance.

1. [MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines](#)

Elective 3: Underground Coal

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This unit is about adhering to health, safety and environmental guidelines in Underground Coal Mines (UCM) during execution of various tasks, operations and maintenance.

1. [MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines](#)

Options(Not mandatory):

Option : Underground Installation

This unit is about complying with unique requirements for installations, operations and maintenance (O&M) of electrical systems in U/G Mines.

1. [MIN/N3104: Comply with unique requirements for installation and operations and maintenance \(O&M\) of electrical system in U/G Mines](#)

Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Engineering Services
Occupation	Electrical Services
Country	India
NSQF Level	4.0
Credits	20
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7411.0200
Minimum Educational Qualification & Experience	12th grade Pass (or equivalent) OR 11th grade pass (with 1.5 year relevant experience in Mining Industry) OR 10th grade pass (with 3 years relevant experience in Mining Industry) OR Previous relevant Qualification of NSQF Level (3.0): (3 years relevant experience)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	Valid gas testing certificate (only for underground mines)

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Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	06/05/2028
NSQC Approval Date	08/05/2025
Version	3.0
Reference code on NQR	QG-04-MG-04060-2025-V2-SCMS
NQR Version	1.0

Remarks:

Total Notional Duration: 420 Hours; Additional 60 hours for Optional NOS= Domain Theory Duration: 90 Hours; Additional 20 hours for Optional NOS Theory Duration+ Domain Practical Duration: 300 Hours; Additional 40 hours for Optional NOS Practical Duration+ Employability Skills Duration: 30 Hours

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MIN/N3101: Identify work requirements, related processes and arrange equipment for electrical system

Description

This unit is about identifying the work and the process requirements of electrical system operation, upkeep and maintenance and the activities & equipment associated with the process to complete the job requirement.

Scope

The scope covers the following :

- Identify the work and process requirements
- Arrange the electrical equipment/machinery/materials to conduct the processes

Elements and Performance Criteria

Identify the work and process requirements

To be competent, the user/individual on the job must be able to:

- PC1.** interpret all the work instructions/ work orders/ work related documents given by the supervisor to understand the requirements for electrical substations/ electrical equipment/ electrical wiring or fixtures
- PC2.** interpret the required work output
- PC3.** identify the safety requirement for mines as per CEA regulation 2010
- PC4.** interpret various electrical layout and discuss with supervisor, in case of any doubt
- PC5.** identify various type of fire protection system and its uses
- PC6.** interpret schematic line diagram of substation and associated installations
- PC7.** interpret the specifications of various systems within electrical substations/ transmission of electricity/operation of electrical equipment/ electrical wiring or fixtures as mentioned in the work instructions/ SOP/ Control Diagrams
- PC8.** prepare/follow sketches/blueprints to determine the location of wiring or equipment ensuring conformance to safety codes

Arrange the electrical equipment/machinery/materials to conduct the processes

To be competent, the user/individual on the job must be able to:

- PC9.** arrange the resources as per the specifications in the work instructions for installation of electrical substations/equipment/ wiring or fixtures
- PC10.** arrange various tools, equipment, measuring apparatus required to perform the job
- PC11.** ensure that the required electrical equipment are in proper working condition
- PC12.** check test equipment (Volt-Ohm Meter) on a known live source of same rating to ensure it works correctly

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** risk and impact of not following defined procedures/work instructions
- KU2.** cost of equipment and loss for the company that results from damage of equipment
- KU3.** implications of delays in the process
- KU4.** different types of mines and detail of the mine one is working in
- KU5.** code of practice in specific areas of the mine
- KU6.** about shot-firing / blasting related safety regulations including taking shelter during blasting
- KU7.** duties of workmen under the Mines Act-1952
- KU8.** provision of compensation and working hours, leaves, etc. as per Mines Act-1952
- KU9.** the outcome of violation of safety procedures
- KU10.** different types of electrical systems and their specifications
- KU11.** different types of electrical requirements in the mine
- KU12.** processes like procurement, store management, inventory management, quality management, etc.
- KU13.** key contact points for query resolution
- KU14.** different units of measurement related to electricity like Watt (for power), Ampere (for current), Ohm (for resistance), Volt (for potential difference), Hertz (for frequency) etc.
- KU15.** about various symbols for electrical drawings and circuit diagrams for the electrical systems/equipment for installation/maintenance
- KU16.** about different types of tools and tackles used in an electrical job
- KU17.** different types of measuring instruments
- KU18.** statutory provisions under relevant electrical laws and rules prescribed by relevant authority
- KU19.** hazards and safety aspects involved in electrical job and usage of relevant PPEs
- KU20.** basic working principle of equipment like transformers, switchgears, relays etc.
- KU21.** Indian Electricity Rules framework

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS5.** make decisions pertaining to the concerned area of work
- GS6.** plan and organize the work order and tasks
- GS7.** detect problems in day to day tasks
- GS8.** find solutions to problems and resolve them
- GS9.** make decisions in emergency conditions
- GS10.** complete the assigned tasks timely

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Identify the work and process requirements</i>	19	29	-	14
PC1. interpret all the work instructions/ work orders/ work related documents given by the supervisor to understand the requirements for electrical substations/ electrical equipment/ electrical wiring or fixtures	2	4	-	3
PC2. interpret the required work output	3	2	-	2
PC3. identify the safety requirement for mines as per CEA regulation 2010	2	4	-	1
PC4. interpret various electrical layout and discuss with supervisor, in case of any doubt	3	3	-	1
PC5. identify various type of fire protection system and its uses	2	2	-	1
PC6. interpret schematic line diagram of substation and associated installations	2	4	-	2
PC7. interpret the specifications of various systems within electrical substations/ transmission of electricity/operation of electrical equipment/ electrical wiring or fixtures as mentioned in the work instructions/ SOP/ Control Diagrams	3	4	-	2
PC8. prepare/follow sketches/blueprints to determine the location of wiring or equipment ensuring conformance to safety codes	2	6	-	2
<i>Arrange the electrical equipment/machinery/materials to conduct the processes</i>	11	21	-	6
PC9. arrange the resources as per the specifications in the work instructions for installation of electrical substations/equipment/ wiring or fixtures	2	5	-	1
PC10. arrange various tools, equipment, measuring apparatus required to perform the job	3	5	-	2
PC11. ensure that the required electrical equipment are in proper working condition	3	5	-	1



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. check test equipment (Volt-Ohm Meter) on a known live source of same rating to ensure it works correctly	3	6	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3101
NOS Name	Identify work requirements, related processes and arrange equipment for electrical system
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Electrical Services
NSQF Level	4
Credits	3
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

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MIN/N3102: Install the electrical systems/sub-stations and equipment

Description

This NOS is about installing electrical systems/sub-stations and electrical equipment as per the specifications required and industry standards.

Scope

The scope covers the following :

- Installation of the electrical supply system and equipment

Elements and Performance Criteria

Installation of the electrical supply system and equipment

To be competent, the user/individual on the job must be able to:

- PC1.** use various types of electrical tools and measuring instruments for installation of various electrical system and equipment
- PC2.** install the required electrical supply systems including transformers, generators, circuit breakers, isolators, bus bars, etc., as per required specified voltage, current, power, energy, frequency, RPM, etc. by using measuring instruments
- PC3.** install required electrical equipment like motors, fans, lighting fixtures, ACs, heaters, compressors, pumps etc.
- PC4.** follow the procedure of obtaining work permit from competent person
- PC5.** conduct a test to ensure the performance of installed electrical equipment as per the defined specifications
- PC6.** make modifications in the parameters (by selecting the right program from the machine control system), if required, and ensure alignment with the prescribed standards
- PC7.** examine all conductors for adequate current carrying capacity and ensure that joints in conductors are properly soldered
- PC8.** check electrical insulation and protection of all live parts (parts carrying current)
- PC9.** ensure all equipment are flameproof
- PC10.** use special precautionary techniques, personal protective equipment (PPE), insulating and shielding materials and tools while working with energized parts
- PC11.** ensure that hand cords and receptacles are dry when plugging and unplugging power cables
- PC12.** identify the criticality of installation operation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** risk and impact of not following defined procedures/work instructions
- KU2.** the hierarchy for reporting identified problems
- KU3.** cost of equipment and loss for the company that results from damage of equipment

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- KU4.** implications of delays in the process
- KU5.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to electricity related operations
- KU6.** benching in quarries, dressing of overhangs, undercuts, fencing
- KU7.** importance of safety in the vicinity of machinery
- KU8.** the electrical layout and the plan showing the position of all fixed apparatus and conductors therein (provisions of Indian Electricity Act, Mines Act, and other regulations)
- KU9.** technical concepts of different types of circuit breakers, switchgear, and their controls
- KU10.** technical concepts of various sensors, special wiring requirements, relays and single-phase preventers, step down system 550/110/9V (Volt) intrinsically safe low voltage equipment, etc.
- KU11.** manufacturer recommendations for troubleshooting and maintenance
- KU12.** the process of laying cables, cable jointing, vulcanizing of trailing cable
- KU13.** allowed voltage restrictions for hand-held portable apparatus (e.g. not exceeding 125 volts)
- KU14.** about maximum stresses, factors of safety and connection with earth-as per the Indian Electricity Rules (IER)
- KU15.** importance of the provision made for immediate and automatic discharge of every static condenser on disconnection of supply
- KU16.** provisions applicable for medium, high, or extra-high voltage installations as per section 51 of IER
- KU17.** sketches and engineering drawings for the installation of electrical systems
- KU18.** different types of electrical tools and machinery
- KU19.** hazards and safety aspects involved and usage of relevant PPEs
- KU20.** electrical defects and how they are generated and can be prevented
- KU21.** how to work with flameproof equipment
- KU22.** about clearance between equipment as per CEA (Central Electricity Authority) Regulations 2010
- KU23.** earthing system for underground mines (Restricted neutral earthing, Neutral Grounding Resistors (NGR), etc.)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS5.** make decisions pertaining to the concerned area of work
- GS6.** plan and organize the work order and tasks
- GS7.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS8.** detect problems in day to day tasks



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- GS9.** find solutions to problems and resolve them
- GS10.** make decisions in emergency conditions
- GS11.** complete the assigned tasks timely

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Installation of the electrical supply system and equipment</i>	30	50	-	20
PC1. use various types of electrical tools and measuring instruments for installation of various electrical system and equipment	3	5	-	1
PC2. install the required electrical supply systems including transformers, generators, circuit breakers, isolators, bus bars, etc., as per required specified voltage, current, power, energy, frequency, RPM, etc. by using measuring instruments	4	6	-	2
PC3. install required electrical equipment like motors, fans, lighting fixtures, ACs, heaters, compressors, pumps etc.	2	4	-	2
PC4. follow the procedure of obtaining work permit from competent person	2	2	-	1
PC5. conduct a test to ensure the performance of installed electrical equipment as per the defined specifications	2	6	-	1
PC6. make modifications in the parameters (by selecting the right program from the machine control system), if required, and ensure alignment with the prescribed standards	3	6	-	2
PC7. examine all conductors for adequate current carrying capacity and ensure that joints in conductors are properly soldered	2	4	-	1
PC8. check electrical insulation and protection of all live parts (parts carrying current)	2	6	-	2
PC9. ensure all equipment are flameproof	3	3	-	2
PC10. use special precautionary techniques, personal protective equipment (PPE), insulating and shielding materials and tools while working with energized parts	3	4	-	2



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. ensure that hand cords and receptacles are dry when plugging and unplugging power cables	3	2	-	2
PC12. identify the criticality of installation operation	1	2	-	2
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3102
NOS Name	Install the electrical systems/sub-stations and equipment
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Electrical Services
NSQF Level	4
Credits	4
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

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MIN/N3103: Operate and maintain electrical systems/substation and equipment

Description

This unit is about operations and maintenance (O&M) of electrical systems/ substations and electrical equipment.

Scope

The scope covers the following :

- Perform electrical operations
- Maintenance procedures

Elements and Performance Criteria

Perform electrical operations

To be competent, the user/individual on the job must be able to:

- PC1.** ensure that the electrical circuit is locked and tagged properly to make it safe to work, before starting the work
- PC2.** operate various electrical equipment such as motor, pump, fan etc.
- PC3.** maintain the records of electrical job related documents as per the statutory requirements
- PC4.** run the installed electrical equipment in the substation to generate and distribute the electricity to the entire mine area with back-ups and redundancies

Maintenance procedures

To be competent, the user/individual on the job must be able to:

- PC5.** follow the procedure to discharge electrical circuit before repair and maintenance
- PC6.** follow the maintenance schedule given by electrical supervisors/engineers
- PC7.** repair and maintain the different electrical equipment as per manufacturer's guidelines/SOPs and statutory requirements (if any)
- PC8.** carry out preventive and break down maintenance for generators, transformers, circuit breakers, isolators, bus bars, control panels, switchboards, wiring, protective relays etc.
- PC9.** inspect electrical equipment to identify electrical risks, hazards, defects or the need for adjustment /repair

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** the hierarchy for reporting identified problems
- KU4.** cost of equipment and loss for the company that results from damage of equipment

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- KU5.** implications of delays in the process
- KU6.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to electricity relevant operations
- KU7.** basic principles of electrical energy, and O&M of electrical systems
- KU8.** about different types of tools and electrical equipment
- KU9.** electrical hazards and its safety aspects
- KU10.** statutory requirements for respective electrical systems
- KU11.** causes and prevention of electrical defects/malfunctions
- KU12.** working of electrical systems and machines
- KU13.** about Diesel Generator (DG), mobile lighting equipment, high mast etc.
- KU14.** installation and handling procedures of safety devices
- KU15.** use of PLC, RLC, ECM etc. (Logic controls)
- KU16.** the process of illumination survey and its standards
- KU17.** electrical defects and how they are generated and can be prevented
- KU18.** the interlocking of equipment as per Central Electricity Authority Regulations
- KU19.** importance of sensitization towards different genders and persons with disabilities (PWD)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS5.** make decisions pertaining to the concerned area of work
- GS6.** plan and organize the work order and tasks
- GS7.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS8.** detect problems in day to day tasks
- GS9.** find solutions to problems and resolve them
- GS10.** make decisions in emergency conditions
- GS11.** complete the assigned tasks timely

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform electrical operations</i>	15	24	-	8
PC1. ensure that the electrical circuit is locked and tagged properly to make it safe to work, before starting the work	4	6	-	1
PC2. operate various electrical equipment such as motor, pump, fan etc.	4	6	-	2
PC3. maintain the records of electrical job related documents as per the statutory requirements	3	6	-	2
PC4. run the installed electrical equipment in the substation to generate and distribute the electricity to the entire mine area with back-ups and redundancies	4	6	-	3
<i>Maintenance procedures</i>	15	26	-	12
PC5. follow the procedure to discharge electrical circuit before repair and maintenance	3	4	-	1
PC6. follow the maintenance schedule given by electrical supervisors/engineers	3	4	-	3
PC7. repair and maintain the different electrical equipment as per manufacturer's guidelines/SOPs and statutory requirements (if any)	3	6	-	3
PC8. carry out preventive and break down maintenance for generators, transformers, circuit breakers, isolators, bus bars, control panels, switchboards, wiring, protective relays etc.	3	6	-	2
PC9. inspect electrical equipment to identify electrical risks, hazards, defects or the need for adjustment /repair	3	6	-	3
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3103
NOS Name	Operate and maintain electrical systems/substation and equipment
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Electrical Services
NSQF Level	4
Credits	4
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/09/2026
NSQC Clearance Date	12/03/2026

Qualification Pack

MIN/N1702: Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines

Description

This unit is about adhering to health, safety and environmental guidelines in Underground Metalliferous Mines (UMM) during the execution of various tasks and operations and maintenance.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work
- PC2.** comply with safety, health and security-related regulations/guidelines at the mine e.g. follow Standard Operating Procedure (SOP) for material handling in underground (U/G) mine
- PC3.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC4.** check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area
- PC5.** take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment
- PC6.** use self-rescue apparatus, appropriately when required
- PC7.** follow appropriate emergency response procedure during emergency such as fire, water inrush, fall of ground etc.
- PC8.** follow precautions against U/G electrical appliances
- PC9.** follow appropriate Standard Operating Procedure while working near any isolated and sealed off area of the mine
- PC10.** follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire
- PC11.** follow the manufacturer's instructions for care and safe operation of mine machinery and equipment
- PC12.** identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face
- PC13.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC14.** follow gas detecting alarm signal on leakage of inflammable gases and laid out procedure to ensure safety
- PC15.** follow process for reporting any unsafe act/condition in work area to the concerned person

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- PC16.** use underground mine communication system
- PC17.** ensure positive isolation near the work place if applicable
- PC18.** use appropriate PPE as per the requirement
- PC19.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC20.** disinfect the machine/tools before and after work/task
- PC21.** maintain hygiene at the work site
- PC22.** report any symptoms of illness to the shift-incharge
- PC23.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC24.** identify the environmental impact of mining related operations and follow steps to reduce those impact
- PC25.** follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery
- PC26.** ensure that the stowing practices produce minimum disturbance to the surface
- PC27.** ensure that the subgrade ore is carried out to surface and stacked separately at the earmarked place
- PC28.** ensure the productivity of the machine for material/fuel conservation
- PC29.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC30.** follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU2.** duties and rights of workers
- KU3.** selection process of person for rescue training
- KU4.** about Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR)
- KU5.** isolation and sealed off area of the mine
- KU6.** various types of gases found in the mine and their effect
- KU7.** self-rescue apparatus and their uses
- KU8.** provision of medical examination (IME & PME) of person employed as per Mines Rules 1955
- KU9.** importance of first aid and hygiene
- KU10.** importance of blind spot survey/training
- KU11.** about different types of machinery used in U/G mines
- KU12.** different types of supporting systems used in U/G mines as per SSP and SSR
- KU13.** about precautions to be taken when handling heavy equipment
- KU14.** various problems/incidents likely to occur
- KU15.** role of Internal Safety Organization, safety committee, workman's inspector and DGMS

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- KU16.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU17.** common sources of pollution in the mines and ways to minimize it
- KU18.** various types of fire extinguishers
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, Gas Detector, Safety Lamp, Self-Contained Breathing Apparatus, gum boots, ear plugs, Face Mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization as per DGMS guideline
- KU22.** mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** Take-5 (Personal Risk Assessment) training (DGMS Tech. circulars 2/2014)
- KU25.** rules and regulations for safety and security while handling hazardous materials
- KU26.** safety appliances and rescue equipment
- KU27.** importance and use of various communication system used in UG mines
- KU28.** importance of positive isolation at working site
- KU29.** safety and occupational health policy of organisation
- KU30.** six directional hazard identification process
- KU31.** basic personal and workplace hygiene
- KU32.** importance of FAB (Fresh Air Base)
- KU33.** basic provisions in Mines Creche Rules, 1966 (MCR) for females employed in the mines
- KU34.** the role and responsibilities of rescue room and rescue station and how to contact them in case of emergency
- KU35.** importance of taking shelter at the miner's station during blasting operation
- KU36.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU37.** importance of following infection control policies, '5-S' practices, and waste management
- KU38.** importance of water/material/energy conservation and management
- KU39.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read basic English language
- GS4.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS5.** use the digital information from machine
- GS6.** use basic applications of a computer
- GS7.** plan and organize the work order and tasks
- GS8.** use reasoning skills to identify and resolve fundamental problems

Qualification Pack

GS9. complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	24	42	-	17
PC1. undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work	-	2	-	2
PC2. comply with safety, health and security-related regulations/guidelines at the mine e.g. follow Standard Operating Procedure (SOP) for material handling in underground (U/G) mine	1	3	-	-
PC3. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	4	-	1
PC4. check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area	1	1	-	-
PC5. take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment	1	2	-	1
PC6. use self-rescue apparatus, appropriately when required	-	4	-	1
PC7. follow appropriate emergency response procedure during emergency such as fire, water inrush, fall of ground etc.	1	2	-	1
PC8. follow precautions against U/G electrical appliances	2	2	-	2
PC9. follow appropriate Standard Operating Procedure while working near any isolated and sealed off area of the mine	2	2	-	1
PC10. follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire	2	4	-	-
PC11. follow the manufacturer's instructions for care and safe operation of mine machinery and equipment	1	2	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face	1	1	-	-
PC13. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	-	-	-
PC14. follow gas detecting alarm signal on leakage of inflammable gases and laid out procedure to ensure safety	1	2	-	1
PC15. follow process for reporting any unsafe act/condition in work area to the concerned person	-	2	-	1
PC16. use underground mine communication system	-	1	-	-
PC17. ensure positive isolation near the work place if applicable	1	1	-	1
PC18. use appropriate PPE as per the requirement	2	4	-	2
PC19. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	1	-	1
PC20. disinfect the machine/tools before and after work/task	1	1	-	-
PC21. maintain hygiene at the work site	1	1	-	1
PC22. report any symptoms of illness to the shift-incharge	1	-	-	-
PC23. identify six directional hazards at workplace and take decisions accordingly	2	-	-	-
<i>Follow environmental guidelines</i>	6	8	-	3
PC24. identify the environmental impact of mining related operations and follow steps to reduce those impact	1	2	-	-
PC25. follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery	-	1	-	-
PC26. ensure that the stowing practices produce minimum disturbance to the surface	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure that the subgrade ore is carried out to surface and stacked separately at the earmarked place	1	-	-	1
PC28. ensure the productivity of the machine for material/fuel conservation	1	1	-	-
PC29. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	1	-	1
PC30. follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1702
NOS Name	Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Underground, Specialized Operations, Electrical Services, Mechanical Services
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

MIN/N1703: Follow Health, Safety, and Environmental Guidelines for opencast mines

Description

This unit is about adhering to health, safety and environmental guidelines at the Open Cast Mines while executing various tasks, maintenance and operations.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow Environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** comply with safety, health, and security-related regulations/guidelines at the opencast mine
- PC2.** follow the safety instructions given by the workman's inspector
- PC3.** follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.
- PC4.** take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations
- PC5.** follow appropriate Safe Operating Procedure (SOP) while dealing with explosives
- PC6.** respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility
- PC7.** provide first aid to an injured person
- PC8.** operate various types of fire extinguishers to control different types of fire at a worksite when required
- PC9.** use appropriate PPE as per the requirement
- PC10.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC11.** disinfect the machine/tools before and after work/task
- PC12.** maintain hygiene at the work site
- PC13.** report any symptoms of illness to the shift-incharge

Follow Environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC14.** identify the environmental impact of related opencast mining operations
- PC15.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC16.** ensure not to mix topsoil with waste in day to day tasks
- PC17.** ensure that HEMM is washed at the designated location

Qualification Pack

- PC18.** ensure the productivity of the machine for material/fuel conservation
- PC19.** follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)
- PC20.** assist supervisor for reducing environmental impact caused due to related mining operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** about various environmental awareness program related to mining, organized by the various government bodies/company
- KU2.** safety guidelines specified by Directorate General of Mine Safety (DGMS)
- KU3.** how to identify six-directional hazards
- KU4.** basic mining terminologies and definitions
- KU5.** importance of blind spot survey/training
- KU6.** means of access and egress from the mines, location of workshop, haul roads and working face including dump yards
- KU7.** duties of workers under The Mines act-1952
- KU8.** working hours and accident compensation as per The Mines act-1952
- KU9.** about precautions to be taken when handling heavy equipment
- KU10.** various problems/incidents likely to occur
- KU11.** hierarchy of the reporting
- KU12.** machine operation, condition of the machine and worksite
- KU13.** proper documents specific to the machine
- KU14.** role of workmen inspector, safety committee and internal safety organization
- KU15.** the process of top soil removal and management
- KU16.** mine sump and pumping system of the mines
- KU17.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU18.** common sources of pollution in the mines and ways to minimize it
- KU19.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, gas detector, safety lamp, self-contained breathing apparatus, gum boots, ear plugs, face mask, etc.
- KU20.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU21.** emergency response /disaster management plan prepared by the organization
- KU22.** signages, mining area-specific signs, and other safety and emergency signals
- KU23.** the outcome of violation of safety procedures
- KU24.** basic personal and workplace hygiene
- KU25.** importance of sensitization towards different genders and PWD (Persons with Disabilities)

Generic Skills (GS)

User/individual on the job needs to know how to:



Skill Council For Mining Sector



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- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS4.** plan and organize the work order and jobs
- GS5.** use reasoning skills to identify and resolve fundamental problems
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	20	34	-	15
PC1. comply with safety, health, and security-related regulations/guidelines at the opencast mine	2	2	-	1
PC2. follow the safety instructions given by the workman's inspector	1	3	-	1
PC3. follow adequate safety while working at haul roads, heights, overburden dumps, sump area, stockyard, near moving parts, etc.	3	3	-	1
PC4. take safety precautions while working on sites (sub-station, workshop etc.), with equipment, and conducting welding and cutting operations	1	3	-	1
PC5. follow appropriate Safe Operating Procedure (SOP) while dealing with explosives	2	3	-	2
PC6. respond promptly and appropriately to an accident/ incident or an emergency situation, within limits of the role and responsibility	2	3	-	1
PC7. provide first aid to an injured person	2	3	-	1
PC8. operate various types of fire extinguishers to control different types of fire at a worksite when required	1	3	-	1
PC9. use appropriate PPE as per the requirement	2	4	-	2
PC10. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	1	2	-	1
PC11. disinfect the machine/tools before and after work/task	1	2	-	1
PC12. maintain hygiene at the work site	1	1	-	1
PC13. report any symptoms of illness to the shift-incharge	1	2	-	1
<i>Follow Environmental guidelines</i>	10	16	-	5

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. identify the environmental impact of related opencast mining operations	2	2	-	1
PC15. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	1	2	-	1
PC16. ensure not to mix topsoil with waste in day to day tasks	2	2	-	1
PC17. ensure that HEMM is washed at the designated location	2	2	-	1
PC18. ensure the productivity of the machine for material/fuel conservation	1	3	-	-
PC19. follow the mineral conservation practices specified by the organization in accordance with MCDR-2017 (Mineral Conservation and Development Rules)	1	3	-	1
PC20. assist supervisor for reducing environmental impact caused due to related mining operations	1	2	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1703
NOS Name	Follow Health, Safety, and Environmental Guidelines for opencast mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, , Exploration, Mineral Estimation, Planning, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Opencast, Specialized Operations, Electrical Services, Mechanical Services, Instrumentation and Control Systems, Ore Processing
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	11/03/2026
Next Review Date	11/03/2029
NSQF Clearance Date	11/03/2026

Qualification Pack

MIN/N1704: Follow Health, Safety, and Environmental guidelines for underground coal mines

Description

This unit is about adhering to health, safety and environmental guidelines in Underground Coal Mines (UCM) during execution of various tasks, operations and maintenance.

Scope

The scope covers the following :

- Follow work-site health and safety measures
- Follow environmental guidelines

Elements and Performance Criteria

Follow work-site health and safety measures

To be competent, the user/individual on the job must be able to:

- PC1.** follow preventive measures against firedamp, whitedamp, blackdamp etc.
- PC2.** use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)
- PC3.** undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work
- PC4.** comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine
- PC5.** ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle
- PC6.** ensure that every instrument, apparatus and equipment are DGMS approved before these are used
- PC7.** ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining
- PC8.** provide first aid to an injured person
- PC9.** follow safety precautions against spontaneous heating of the coal
- PC10.** operate various types of fire extinguishers to control different types of fire at worksite, if required
- PC11.** ensure that no person is traveling/working/staying under unsupported roof
- PC12.** check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area
- PC13.** take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment
- PC14.** use self-rescue apparatus appropriately when required
- PC15.** follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC16.** follow precautions against U/G electrical appliances

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- PC17.** take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports
- PC18.** follow appropriate SOP while working near any isolated and sealed off area of the mine
- PC19.** ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly
- PC20.** take relevant safety precautions during depillaring operation in UCM
- PC21.** follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire
- PC22.** follow the manufacturer's instructions for care and safe operation of mine machinery and equipment
- PC23.** identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face
- PC24.** follow laid out SOP in case of alarm signal for leakage of inflammable gases
- PC25.** follow the process of reporting any unsafe act/condition in the working area to the concerned person
- PC26.** use underground mine communication system
- PC27.** ensure positive isolation near the work place if applicable
- PC28.** use appropriate Personal Protective Equipment (PPE) as per the requirement
- PC29.** maintain hand hygiene by washing hands with alcohol based sanitisers/soap
- PC30.** disinfect the machine/tools before and after work/task
- PC31.** maintain hygiene at the work site
- PC32.** report any symptoms of illness to the shift-incharge
- PC33.** identify six directional hazards at workplace and take decisions accordingly

Follow environmental guidelines

To be competent, the user/individual on the job must be able to:

- PC34.** identify the environmental impact of mining related operations and follow steps to reduce those impact
- PC35.** follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery
- PC36.** ensure that the stowing practices produce minimum disturbance to the surface
- PC37.** ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place
- PC38.** ensure the productivity of the machine for material/fuel conservation
- PC39.** follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines
- PC40.** follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** safety guidelines specified by Directorate General of Mine Safety (DGMS)

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- KU2.** duties and rights of workers
- KU3.** selection process of person for rescue training
- KU4.** about Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR)
- KU5.** isolation and sealed off area of the mine
- KU6.** various types of gases available in the mine and their effects; and their control measures
- KU7.** self-rescue apparatus and their uses
- KU8.** provision of medical examination (Initial Medical Examination (IME) & Periodical Medical Examination (PME)) of a person employed, as per Mines Rules 1955
- KU9.** importance of first aid and hygiene
- KU10.** about different types of machineries used in U/G mines
- KU11.** different types of supporting system used in U/G mines as per SSP and SSR
- KU12.** about precautions to be taken when handling heavy equipment
- KU13.** various problems/incidents likely to occur
- KU14.** role of Internal Safety Organization, safety committee, workman's inspector and DGMS
- KU15.** about mine safety standard including light illumination level, noise levels, dust level, pollutants, etc at the work-site
- KU16.** common sources of pollution in the mines and ways to minimize it
- KU17.** various types of fire extinguishers
- KU18.** safety equipment like safety shoes, safety belt, tight fit clothing, hand gloves, safety goggles, Gas Detector, Safety Lamp, Self-Contained Breathing Apparatus, gum boots, ear plugs, Face Mask, etc.
- KU19.** shot-firing / blasting related safety regulations including taking shelter during blasting
- KU20.** mining area-specific signs, and other safety and emergency signals
- KU21.** the outcome of violation of safety procedures
- KU22.** Take-5 (Personal Risk Assessment) training (DGMS Tech. circulars 2/2014)
- KU23.** hazardous material safety, security rules and regulations
- KU24.** safety appliances and rescue equipment
- KU25.** importance and use of various communication system used in UG mines
- KU26.** importance of positive isolation at working site
- KU27.** safety and occupational health policy of organisation
- KU28.** six directional hazard identification process
- KU29.** importance of blind spot survey/training
- KU30.** basic personal and workplace hygiene
- KU31.** importance of FAB (Fresh Air Base)
- KU32.** basic provisions in Mines Creche Rules, 1966 (MCR) for any females employed in the mines
- KU33.** about basic safety regulations of Coal Mines Regulation, 2017 (CMR)
- KU34.** types of stone dust barrier and its importance
- KU35.** coal dust explosion and its preventive measures
- KU36.** classification of coal mines as per the degree of gassiness of coal seams such as first degree, second degree, and third-degree mines
- KU37.** precautions as per the gassiness of the coal mines

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- KU38.** use of flame safety lamp and its parts
- KU39.** about coal mines occupational disease such as pneumoconiosis or 'black lung' and their preventive measures
- KU40.** Standard of Ventilation as per the section 153 of the CMR 2017
- KU41.** Standard of Lighting as per the section 178 of the CMR 2017
- KU42.** the roles, duties and responsibilities of rescue team members, rescue room and rescue station and how to contact them in case of emergency
- KU43.** the correct steps for conducting any rescue work as per Mine Rescue Rule (MRR)
- KU44.** importance of taking shelter at the miner's station during blasting operation
- KU45.** importance of sensitization towards different genders and persons with disabilities (PWD)
- KU46.** importance of following infection control policies, '5-S' practices, and waste management
- KU47.** importance of water/material/energy conservation and management
- KU48.** importance of SMP and EMP, prepared by the organization, as per DGMS guideline

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** make decisions on the concerned area of work
- GS3.** read basic English language
- GS4.** read and interpret manuals, health, and safety instructions, memos, etc.
- GS5.** use the digital information from a machine
- GS6.** plan and organize the work order and tasks
- GS7.** use basic applications of the computer
- GS8.** use reasoning skills to identify and resolve fundamental problems
- GS9.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow work-site health and safety measures</i>	24	43	-	14
PC1. follow preventive measures against firedamp, whitedamp, blackdamp etc.	1	1	-	-
PC2. use the flame safety lamp for detecting the methane gas as per Standard Operating Procedure (SOP)	-	1	-	-
PC3. undertake "The Take-5 (Personal Risk Assessment)" before commencement of any work	1	1	-	1
PC4. comply with safety, health and security-related regulations/guidelines at the mine e.g. SOP for material handling in underground (U/G) mine	1	1	-	1
PC5. ensure that oil, grease, canvas or other inflammable material are stored in fire-proof receptacle	-	1	-	-
PC6. ensure that every instrument, apparatus and equipment are DGMS approved before these are used	1	1	-	-
PC7. ensure that Armoured face conveyor (AFC) and chocks must be kept in a straight line for every cycle of operations and tightened up to the setting pressure while keeping it in full contact with the roof, applicable for longwall mining	3	3	-	-
PC8. provide first aid to an injured person	1	2	-	1
PC9. follow safety precautions against spontaneous heating of the coal	1	1	-	-
PC10. operate various types of fire extinguishers to control different types of fire at worksite, if required	-	2	-	-
PC11. ensure that no person is traveling/working/staying under unsupported roof	1	-	-	-
PC12. check that roof supporting is as per Systematic Support Plan (SSP) and approved Systematic Support Rules (SSR) while undertaking work in an area	-	2	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. take precaution against occupational health hazards (like dust, water, mine gases etc.) due to U/G working environment	-	1	-	1
PC14. use self-rescue apparatus appropriately when required	-	2	-	-
PC15. follow Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	1
PC16. follow precautions against U/G electrical appliances	2	2	-	1
PC17. take proper care against damage and accidents while loading, transporting, dismantling and erecting of roof supports	1	2	-	-
PC18. follow appropriate SOP while working near any isolated and sealed off area of the mine	1	1	-	1
PC19. ensure that the roof and the sidewalls of the mine face (or newly exposed area of the mines) have been scaled/dressed properly	1	1	-	-
PC20. take relevant safety precautions during depillaring operation in UCM	-	2	-	1
PC21. follow appropriate safety practices while traveling on U/G haul roads, incase of post blast fumes and misfire	1	1	-	-
PC22. follow the manufacturer's instructions for care and safe operation of mine machinery and equipment	-	1	-	1
PC23. identify the working ventilation district from line diagram of ventilation circuit to direct air to the working face	-	1	-	1
PC24. follow laid out SOP in case of alarm signal for leakage of inflammable gases	1	1	-	-
PC25. follow the process of reporting any unsafe act/condition in the working area to the concerned person	-	1	-	1
PC26. use underground mine communication system	-	1	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC27. ensure positive isolation near the work place if applicable	1	1	-	-
PC28. use appropriate Personal Protective Equipment (PPE) as per the requirement	2	2	-	1
PC29. maintain hand hygiene by washing hands with alcohol based sanitisers/soap	-	1	-	-
PC30. disinfect the machine/tools before and after work/task	1	1	-	-
PC31. maintain hygiene at the work site	1	1	-	-
PC32. report any symptoms of illness to the shift-incharge	1	1	-	-
PC33. identify six directional hazards at workplace and take decisions accordingly	-	1	-	1
<i>Follow environmental guidelines</i>	6	7	-	6
PC34. identify the environmental impact of mining related operations and follow steps to reduce those impact	1	1	-	1
PC35. follow the mineral conservation practices in U/G mining operations to achieve optimum ore or mineral recovery	1	1	-	1
PC36. ensure that the stowing practices produce minimum disturbance to the surface	1	1	-	-
PC37. ensure that the subgrade coal is carried out to surface and stacked separately at the earmarked place	1	1	-	1
PC38. ensure the productivity of the machine for material/fuel conservation	1	1	-	1
PC39. follow the process for collecting, storing and disposing of the hazardous material and waste (like used oil, lubricant, battery, etc.) in compliance with worksite guidelines	-	1	-	1
PC40. follow the "5-S" practice at work site like cleaning oil from ground (to avoid soil from getting damaged), etc.	1	1	-	1



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1704
NOS Name	Follow Health, Safety, and Environmental guidelines for underground coal mines
Sector	Mining
Sub-Sector	Mining Operation
Occupation	HSE Functions, Mine Surveying, Drilling/Cutting, Shot firing/Blasting, Loading and Hauling - Underground, Specialized Operations, Electrical Services, Mechanical Services
NSQF Level	3.5
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	25/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

MIN/N3104: Comply with unique requirements for installation and operations and maintenance (O&M) of electrical system in U/G Mines

Description

This unit is about complying with unique requirements for installations, operations and maintenance (O&M) of electrical systems in Underground (U/G) Mines.

Scope

The scope covers the following :

- Comply with unique requirements for U/G Mine working

Elements and Performance Criteria

Comply with unique requirements for U/G Mine working

To be competent, the user/individual on the job must be able to:

- PC1.** conduct earthing for U/G mines considering all safety precautions
- PC2.** repair and set earth leakage current and time in earth leakage relay
- PC3.** locate the right area for installation of sub-station, equipment in U/G mines ensuring safety and ease of maintenance at the selected area
- PC4.** conduct installation, O&M of signalling and telecom equipment as per intrinsically safe requirements
- PC5.** prepare the underground electrical plan/layout
- PC6.** maintain electrical lamp room and ensure that all lamps are cleaned, refitted and refilled in a separate room
- PC7.** install various safety equipment used in U/G mines
- PC8.** install electric lighting in gassy mines as per CMR (Coal Mines Regulations)
- PC9.** follow all safety standards relevant to electrical layout, earthing system, voltage limit, armored cables and flexible cables as per electricity rules and DGMS guidelines
- PC10.** use portable hand-lamps in U/G mines
- PC11.** extend the relevant electrical cables as per the face development while considering all relevant safety precautions
- PC12.** isolate transformer and switch gears and other electrical equipment
- PC13.** ensure proper lighting is available before commencement of any electrical work
- PC14.** use circuit for the remote control or electric interlocking of apparatus in U/G mines
- PC15.** ensure the electrical circuit is locked and tagged before any installation and O&M
- PC16.** ensure that all power circuits and electrical equipment is deenergized before work is performed, except for troubleshooting or testing
- PC17.** remove electrical equipment from service in case any potentially dangerous condition is identified
- PC18.** check safety circuits of mining equipment like drilling machines, locomotives, etc.
- PC19.** ensure equipment are flameproof while working in gassy mines

Qualification Pack

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** job-specific documents e.g. daily maintenance checklist and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** cost of equipment and loss for the company that results from damage of equipment
- KU4.** implications of delays in the process
- KU5.** safety guidelines specified by Directorate General of Mines Safety (DGMS) specific to U/G electricity relevant operations
- KU6.** about tramways and siding, haulage rooms, winding rooms, boilers, electrical gears, etc.
- KU7.** the outcome of violation of safety procedures
- KU8.** the proper earthing procedures
- KU9.** types of cables, used in mines
- KU10.** precautions to be taken for the gases that exist in U/G mines
- KU11.** use of flameproof equipment/enclosures
- KU12.** earthing of neutral points
- KU13.** setting of earth leakage and overload relays
- KU14.** permissible voltage level for various applications in the underground as per CEA Regulations 2010
- KU15.** how to use portable hand lamps in U/G mines
- KU16.** working principle and use of Neutral Grounding Resistors (NGR) in underground mines
- KU17.** how to use the circuit for the remote control or electric interlocking of apparatus in U/G mines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** note down observations (if any)
- GS2.** read and interpret symbols and readings
- GS3.** read information documents
- GS4.** effectively communicate, listen and comprehend the information given by various sources about the site
- GS5.** make decisions pertaining to the concerned area of work
- GS6.** plan and organize the work order and tasks
- GS7.** organize all operation and service manuals so that sorting/ accessing information is easy
- GS8.** detect problems in day to day tasks
- GS9.** find solutions to problems and resolve them
- GS10.** make decisions in emergency conditions
- GS11.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Comply with unique requirements for U/G Mine working</i>	30	50	-	20
PC1. conduct earthing for U/G mines considering all safety precautions	2	4	-	1
PC2. repair and set earth leakage current and time in earth leakage relay	2	4	-	1
PC3. locate the right area for installation of sub-station, equipment in U/G mines ensuring safety and ease of maintenance at the selected area	2	4	-	2
PC4. conduct installation, O&M of signalling and telecom equipment as per intrinsically safe requirements	2	2	-	1
PC5. prepare the underground electrical plan/layout	2	4	-	1
PC6. maintain electrical lamp room and ensure that all lamps are cleaned, refitted and refilled in a separate room	3	3	-	1
PC7. install various safety equipment used in U/G mines	2	2	-	1
PC8. install electric lighting in gassy mines as per CMR (Coal Mines Regulations)	2	3	-	1
PC9. follow all safety standards relevant to electrical layout, earthing system, voltage limit, armored cables and flexible cables as per electricity rules and DGMS guidelines	1	2	-	1
PC10. use portable hand-lamps in U/G mines	1	2	-	1
PC11. extend the relevant electrical cables as per the face development while considering all relevant safety precautions	2	3	-	1
PC12. isolate transformer and switch gears and other electrical equipment	1	3	-	1
PC13. ensure proper lighting is available before commencement of any electrical work	1	3	-	1

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. use circuit for the remote control or electric interlocking of apparatus in U/G mines	1	1	-	1
PC15. ensure the electrical circuit is locked and tagged before any installation and O&M	1	2	-	1
PC16. ensure that all power circuits and electrical equipment is deenergized before work is performed, except for troubleshooting or testing	1	2	-	1
PC17. remove electrical equipment from service in case any potentially dangerous condition is identified	2	2	-	1
PC18. check safety circuits of mining equipment like drilling machines, locomotives, etc.	1	2	-	1
PC19. ensure equipment are flameproof while working in gassy mines	1	2	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3104
NOS Name	Comply with unique requirements for installation and operations and maintenance (O&M) of electrical system in U/G Mines
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Electrical Services
NSQF Level	4
Credits	2
Version	2.0
Last Reviewed Date	08/05/2025
Next Review Date	06/05/2028
NSQC Clearance Date	08/05/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SCMS will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SCMS.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).

Qualification Pack

5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.

6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.

7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N3101. Identify work requirements, related processes and arrange equipment for electrical system	30	50	0	20	100	20
MIN/N3102. Install the electrical systems/sub-stations and equipment	30	50	0	20	100	30
MIN/N3103. Operate and maintain electrical systems/substation and equipment	30	50	0	20	100	20
DGT/VSQ/N0101. Employability Skills (30 Hours)	20	30	-	-	50	10
Total	110	180	-	60	350	80

Elective: 1 Underground Metal

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1702.Follow Health, Safety and Environmental guidelines for Underground Metalliferous Mines	30	50	-	20	100	20
Total	30	50	-	20	100	20

Elective: 2 Opencast

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1703.Follow Health, Safety, and Environmental Guidelines for opencast mines	30	50	-	20	100	20
Total	30	50	-	20	100	20

Elective: 3 Underground Coal

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N1704.Follow Health, Safety, and Environmental guidelines for underground coal mines	30	50	-	20	100	20
Total	30	50	-	20	100	20

Optional: 1 Underground Installation



Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N3104. Comply with unique requirements for installation and operations and maintenance (O&M) of electrical system in U/G Mines	30	50	0	20	100	20
Total	30	50	-	20	100	20

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
QP	Qualification Pack
NSQF	National Skills Qualifications Framework
TVET	Technical and Vocational Education and Training
SCMS	Skill Council for Mining Sector
ROPS	Roll-Over Protection Structure
FOPS	Falling Object Protective Structure
PC	Performance Criteria
HEMM	Heavy Earth Moving Machinery
OEM	Original Equipment Manufacturer
DGMS	Directorate General of Mining Safety
MMR	Metalliferous Mines Regulations
CMR	Coal Mines Regulations
SOP	Standard Operating Practice
MVTR	Mine Vocational Training Rules
MRR	Mines Rescue Rules
HSE	health, safety & environment
OB	Over Burden
ERP	Enterprise resource planning
MCDR	Mineral Conservation and Development Rules

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	A sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	a sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in the industry.
Occupational Standards	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.

Qualification Pack

Qualification Pack	QP comprises the set of OS, together with the educational, training, and other criteria required to perform a job role. A QP is assigned a unique qualifications pack role
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.